

U3 Wire Antennas and Radiation Parameters

Types of Antennas

- i) Hertzian Dipole $\Rightarrow \lambda < \lambda_{50}$
- ii) Short Dipole $\Rightarrow \frac{\lambda}{50} < \lambda < \frac{\lambda}{10}$
- iii) Basic Dipole (Half Wavelength) $\Rightarrow \frac{\lambda}{2}$ (Max Radiation Efficiency)

Hertzian Dipole

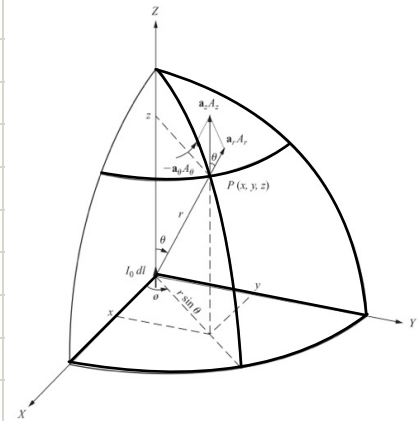
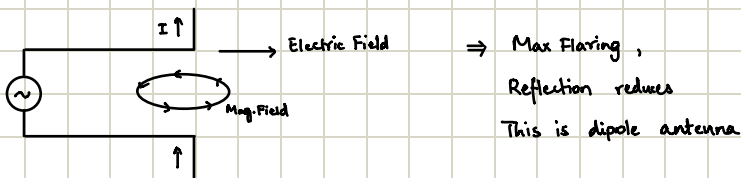
→ Minimum use, Low Radiation Efficiency, Higher Loss, Larger reactive field, Not used commercially

Dipole Antenna Radiation

→ Consider a wire Tx line with $Z_L = \infty$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{1 - \frac{Z_0}{Z_L}}{1 + \frac{Z_0}{Z_L}} = \frac{1 - 0}{1 + 0} = 1 \Rightarrow \text{Max. Reflection}$$

→ Now take an end which is flaring



Radiation Resistance & Radiated Power

- 1) Calculate $\vec{E}$ & $\vec{H}$
- 2) Calculate avg. power density, $\vec{W}_{avg} = \frac{1}{2}(\vec{E} \times \vec{H})$
- 3) Calculate radiated power, $P_{rad} = \int \vec{W}_{avg} \cdot d\vec{s}$
- 4) Calculate radiation resistance, $P_{rad} = \frac{1}{2} I_0^2 R_r \Rightarrow R_r = \frac{2P_{rad}}{I_0^2}$

Infinitesimal dipole

→ Spatial variation of current, $I(z') = \hat{a}_z \cdot I_0$

Radiated fields

→ To find fields radiated by current element by finding A & F, then E & H

$$A(x, y, z) = \frac{\mu}{4\pi} \int_C I_2(x', y', z') \frac{e^{-jkr}}{R} dl'$$

(x, y, z) : observation point

(x', y', z') : source point

R : Distance from any point on source to observation point

C : Path along length of the source

$$I_e(x', y', z') = \hat{a}_z I_0$$

$$(x', y', z') = (0, 0, 0) \Rightarrow R = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2} = \sqrt{x^2 + y^2 + z^2} = r \text{ (constant)}$$

$$A(x, y, z) = \hat{a}_z \frac{\mu I_0}{4\pi r} e^{-jkr} \int_{-\lambda/2}^{\lambda/2} dz' = \hat{a}_z \frac{\mu_0 I_0 \lambda}{4\pi r} e^{-jkr}$$

$$A_r = A_z \cos\theta = \frac{\mu_0 I_0 \lambda}{4\pi r} e^{-jkr} \cdot \cos\theta$$

$$A_\theta = -A_z \sin\theta = \frac{-\mu_0 I_0 \lambda}{4\pi r} e^{-jkr} \cdot \sin\theta$$

$$A_\phi = 0$$

No variations in $\phi \Rightarrow H = \frac{\nabla \times A}{\mu} = \hat{a}_\phi \cdot \frac{1}{\mu r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial}{\partial \theta} A_r \right]$

$H_r = H_\theta = 0$

$H_\phi = j \frac{\kappa I_0 l \sin \theta}{4\pi r} \left(1 + \frac{1}{jkr} \right) e^{-jkr}$

$E_A = -j\omega A - j \frac{1}{\omega \mu \epsilon} \nabla (\nabla \cdot A) = \frac{1}{j\omega \epsilon} \nabla \times H$

$E_r = \frac{\eta I_0 l \cos \theta}{2\pi r^2} \left(1 + \frac{1}{jkr} \right) e^{-jkr}$

$E_\theta = \frac{j\eta \kappa I_0 l \sin \theta}{4\pi r} \left(1 + \frac{1}{jkr} - \frac{1}{(kr)^2} \right) e^{-jkr}$

$E_\phi = 0$

→ Intrinsic Impedance

$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{\kappa}{\omega \epsilon} = 120\pi$

Field Terms

→ Fields decay differently with distance

$\frac{1}{r^3}$: Very close field ; $\frac{1}{r^2}$: Medium distance ; $\frac{1}{r}$: Far field (radiation)

$E_r = \eta \frac{\kappa^2 I_0 l \cos \theta}{2\pi} e^{-jkr} \left(\frac{1}{(kr)^2} + \frac{1}{j(kr)^3} \right)$

$E_\theta = j\eta \frac{\kappa^2 I_0 l \sin \theta}{4\pi} e^{-jkr} \left(\frac{1}{kr} + \frac{1}{j(kr)^2} - \frac{1}{(kr)^3} \right)$

Far field eq's

$E_\theta = j\eta \frac{\kappa I_0 l \sin \theta}{4\pi} \cdot \frac{e^{-jkr}}{r}$
 $H_\phi = j \frac{\kappa I_0 l \sin \theta}{4\pi} \cdot \frac{e^{-jkr}}{r}$
 $\left. \begin{array}{l} E_\theta = \eta H_\phi \\ 120\pi/377 \Omega \text{ for free space} \end{array} \right\}$

→ Waves move outward like spheres

Fields depend on θ

Max radiation @ $\theta = 90^\circ$

Zero radiation @ $\theta = 0^\circ$ & 180°

Near field : $kr \ll 1$

Intermediate field : $kr > 1$

Far field : $kr \gg 1$

Power Density

→ $W_{av} = \frac{1}{2} [E \times H^*]$

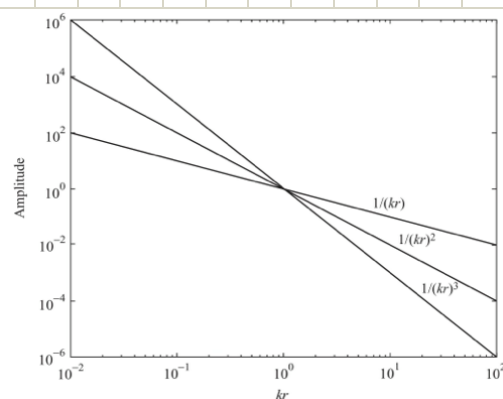
$= \frac{1}{2} \operatorname{Re} \left\{ \hat{a}_r \eta |H_\phi|^2 \right\} = \hat{a}_r \frac{\eta}{2} \left(\frac{\kappa I_0 l}{4\pi} \right)^2 \left(\frac{\sin \theta}{r} \right)^2 = \frac{\eta}{2} |H_\phi|^2$

→ Total power radiated, $P_{rad} = \oint_S W_{av} \cdot d\mathbf{a}$

$= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{\eta}{2} \left(\frac{\kappa I_0 l}{4\pi} \right)^2 \left(\frac{\sin \theta}{r} \right)^2 \cdot r^2 \sin \theta d\theta d\phi$

$= \frac{\eta}{2} \left(\frac{\kappa^2 I_0^2 l^2}{16\pi^2} \right) \cdot 2\pi \cdot \frac{4}{3} = \eta \left(\frac{\kappa^2 I_0^2 l^2}{12\pi} \right)$

Power radiated by Hertzian dipole



EXAMPLE 1.6

Calculate and compare the r and θ components of the electric field intensities at $x = 100$ m, $y = 100$ m, and $z = 100$ m produced by a Hertzian dipole of length $dl = 1$ m kept at the origin, oriented along the z -axis, excited by a current of $i(t) = 1 \times \cos(10\pi \times 10^6 t)$ A, and radiating into free space.

Solution: The frequency of excitation is $\omega = 10\pi \times 10^6$ rad/s, and therefore $f = 5 \times 10^6$ Hz. The dipole is radiating in free space with parameters $\mu = 4\pi \times 10^{-7}$ H/m and $\epsilon = 8.854 \times 10^{-12}$ F/m. Therefore, the impedance of the medium is

$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.854 \times 10^{-12}}} = 376.73 \Omega$

and the propagation constant is

$k = \omega \sqrt{\mu \epsilon} = 10\pi \times 10^6 \sqrt{4\pi \times 10^{-7} \times 8.854 \times 10^{-12}} = 0.1047 \text{ rad/m}$

The distance r between the field point and the dipole (which is at the origin) is

$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{100^2 + 100^2 + 100^2} = 173.2 \text{ m}$

Using the relation $z = r \cos \theta$, we can compute value of θ as

$\theta = \cos^{-1} \left(\frac{z}{r} \right) = \cos^{-1} \left(\frac{100}{173.2} \right) = 54.73^\circ$

Substituting these values in Eqn (1.64)

$E_r = \eta \frac{I_0 dl \cos \theta}{2\pi r} e^{-jkr} \left(1 + \frac{1}{jkr} \right)$
 $= 376.73 \frac{1 \times 1 \times \cos 54.73^\circ}{2\pi \times 173.2} e^{-j0.1047 \times 173.2} \left(1 + \frac{1}{j0.1047 \times 173.2} \right)$
 $= 1.154 \times 10^{-3} (1 - j0.055) e^{-j18.14}$
 $= 1.154 \times 10^{-3} \times 1.002 \angle -3.148^\circ \times 1 \angle 40.65^\circ$
 $= 1.156 \times 10^{-3} \angle 37.51^\circ \text{ V/m}$

The θ -component of the electric field is evaluated using Eqn (1.65)

$E_\theta = j\eta \frac{\kappa I_0 dl \sin \theta}{4\pi r} e^{-jkr} \left(1 + \frac{1}{jkr} - \frac{1}{(kr)^2} \right)$
 $= j376.73 \frac{0.1047 \times 1 \times 1 \times \sin 54.73^\circ}{4\pi \times 173.2} e^{-j18.14} \left(1 + \frac{1}{j18.14} - \frac{1}{18.14^2} \right)$
 $= j0.0148 e^{-j18.14} (1 - j0.055 - 3.04 \times 10^{-3})$
 $= 1 \angle 90^\circ \times 0.0148 \times 1 \angle 40.65^\circ \times 0.9985 \angle -3.158^\circ$
 $= 0.0148 \angle 127.49^\circ \text{ V/m}$

The wavelength of the EM wave is 60 m and, therefore, at a distance of 2.88 λ the θ -component of the electric field is more than 10 times greater than the r -component.

Wire Antennas

- i) Hertzian dipole $\Rightarrow l < \lambda/20$
- ii) Short dipole $\Rightarrow \lambda/20 < l < \lambda/10$
- iii) $\lambda/2$ long dipole $\Rightarrow \lambda/2$
- iv) Any length dipole

short dipole

$$\begin{aligned} H_\phi &= \frac{jkIl\sin\theta e^{-jkr}}{8\pi r} \\ E_\theta &= \frac{jk^2Il\sin\theta e^{-jkr}}{8\pi\omega\epsilon r} \end{aligned} \quad \left. \begin{array}{l} \text{Magnitude of fields are half that of hertzian dipole} \\ \text{Power radiated by short dipole} = \frac{1}{4} \times \text{Power radiated by hertzian dipole} \end{array} \right\}$$

$$W_{av} = \frac{1}{2} |E_\theta| |H_\phi| = \frac{1}{2} \left[\frac{kIl\sin\theta}{8\pi r} \right] \cdot \left[\frac{k^2Il\sin\theta}{8\pi\omega\epsilon r} \right] = \frac{k^3 I^2 l^2 \sin^2\theta}{128\pi^2 r^2 \omega\epsilon} \quad \text{where } E_\theta = \eta H_\phi$$

Power Density

$$\text{Power radiated, } P_{rad}(\text{short}) = \frac{1}{4} P_{rad}(\text{hertzian}) = \frac{\eta k^2 I^2 l^2}{48\pi} = 2.5 k^2 I^2 l^2$$

$$\text{Radiation resistance, } R_{rad} = \frac{P_{rad}}{I_{eff}^2} = \frac{P_{rad}}{I^2/2} = \frac{2 \times \eta k^2 I^2 l^2}{48\pi \times I^2} = \frac{\eta k^2 l^2}{24\pi} = \frac{5}{12\pi} k^2 l^2 = 5k^2 l^2$$

$$k = \omega \sqrt{\mu\epsilon} = \frac{2\pi f}{c} = \frac{2\pi}{\lambda}$$

$$R_{rad} = 5 \times \frac{4\pi^2}{\lambda^2} \times l^2 = \frac{20\pi^2 l^2}{\lambda^2}$$

Half dipole

→ In short dipole → Triangular current

In $\lambda/2$ dipole → Sinusoidal current (current becomes sinusoidal as length increases) (Only in z-direction)

$$\rightarrow I(z') = a_z I_z(z') = \begin{cases} I_0 \cos(kz'), & -\lambda/4 < z' < \lambda/4 \\ 0, & \text{otherwise} \end{cases} \quad \begin{array}{l} \text{Max at } z=0 \\ 0 \text{ at } z=\lambda/4 \text{ \& } -\lambda/4 \end{array}$$

$$\rightarrow \text{Vector potential, } A_z = \frac{\mu}{4\pi} \cdot \frac{e^{-jkr}}{r} \int I(z) e^{jkz \cos\theta} dz$$

$$= \frac{\mu I_0}{2\pi k} \cdot \frac{e^{-jkr}}{r} \cdot \frac{\cos(\frac{\pi}{2} \cos\theta)}{\sin\theta}$$

We can further get H_ϕ & E_θ

$$H_\phi = \frac{jI_0}{2\pi} \cdot \frac{e^{-jkr}}{r} \cdot \frac{\cos(\frac{\pi}{2} \cos\theta)}{\sin\theta}$$

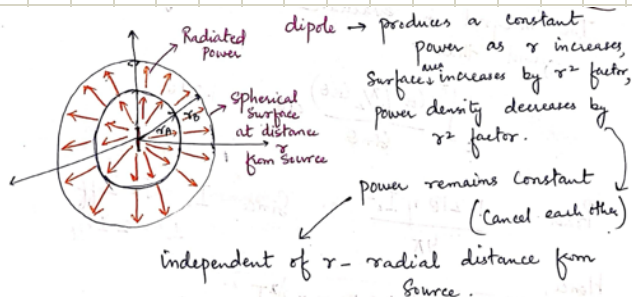
$$E_\theta = \eta H_\phi = \frac{j\eta I_0}{2\pi} \cdot \frac{e^{-jkr}}{r} \cdot \frac{\cos(\frac{\pi}{2} \cos\theta)}{\sin\theta}$$

$$\rightarrow \text{Average Power Density, } W_{av} = \frac{\eta I_0^2}{8\pi^2 r^2} \left(\frac{\cos(\frac{\pi}{2} \cos\theta)}{\sin\theta} \right)^2$$

$$\rightarrow \text{Total Power Radiated, } P_{rad} = \oint_S W_{av} \cdot da = \frac{1.218 \eta I_0^2}{4\pi} = \frac{1.218 \eta}{2\pi} \cdot I_{eff}^2$$

$$\text{In free space, } \eta = 120\pi, \quad P_{rad} = 73 I_{eff}^2$$

$$\rightarrow \text{Radiation Resistance, } R_{rad} = \frac{P_{rad}}{I_{eff}^2} = 73 \Omega$$



Antenna Parameters

i) **Power Density**, Amt. of power per unit area in radiated field

$$S = W_{av} = \frac{P_{rad}}{4\pi r^2} \quad \text{W/m}^2$$

ii) **Power Radiated**, Total power emitted by an antenna into free space

$$P_{rad} = \oint_S W_{av} \cdot da$$

$$= \begin{cases} \eta \left(\frac{\kappa^2 I_0^2 l^2}{12\pi} \right), & \text{Hertzian dipole} \\ \frac{\eta}{4} \left(\frac{\kappa^2 I_0 l^2}{12\pi} \right), & \text{Short dipole} \\ \frac{1.218\eta}{2\pi} \cdot I_{eff}^2, & \text{Half dipole} \end{cases}$$

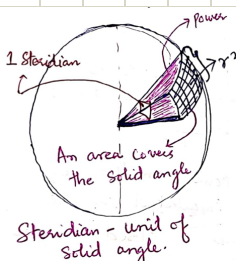
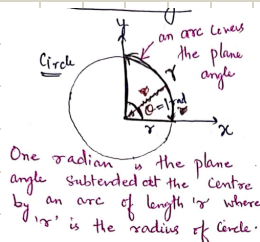
iii) **Radiation Intensity**, Power radiated per unit

$$U = r^2 S$$

For plane angle, Circumference = $2\pi r$

$$\pi \text{ radian} = 180^\circ$$

$$1 \text{ radian} = 57.295^\circ$$



For solid angle, surface area of sphere = $4\pi r^2$

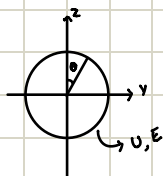
$$4\pi \text{ steradian} = 4\pi r^2$$

$$1 \text{ steradian} = r^2$$

iv) **Radiation Pattern** (Spatial distribution of radiated power)

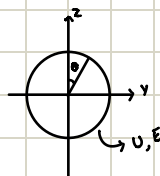
→ 2 orthogonal planes

i) yz plane / θ plane / E plane



Radiation intensity is a circle in yz plane for isotropic antenna

ii) xy plane / ϕ plane / H plane



Radiation intensity is a circle in xy plane for isotropic antenna

v) **HPBW - Half Power Beamwidth**

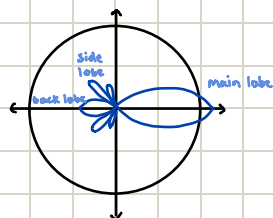
→ Angular separation in which magnitude of radiation pattern decrease by 50% (-3dB) from peak of main beam

$$HPBW = \frac{70\lambda}{D}$$

→ Main lobe : Major part of radiated field (max radiated energy)

Side lobe : Other parts of pattern where radiation is distributed sideways (power wasted)

Back lobe : Lobe exactly opposite to main lobe



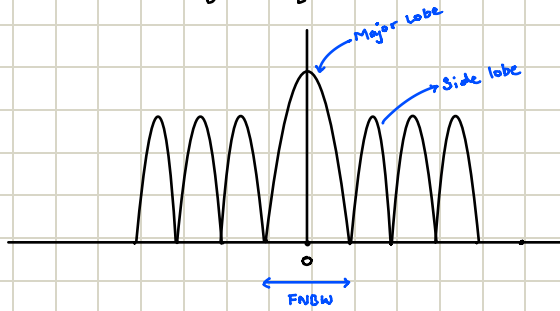
$$1 \text{ dB} = 10 \log \left(\frac{P_{out}}{P_{in}} \right) \Rightarrow 1 \text{ dB} = 10 \log \left(\frac{P_m/2}{P_{in}} \right) = 10 \log \left(\frac{1}{2} \right) = -3 \text{ dB}$$

$$HPBW = \theta_2 - \theta_1$$

FNBW - First Null Beam Width

→ The angular separation b/w first nulls of radiation pattern

$$\text{FNBW} = 2 \cdot \text{HPBW} = 2 \times \frac{70\lambda}{D} = \frac{140\lambda}{D}$$



vi) **Directivity of an Antenna** - Ratio of radiation intensity in a given direction to radiation intensity averaged over all directions
 radiation intensity averaged over all directions = Total power radiated by antenna by 4π

$$D = \frac{U}{U_0} = \frac{U \text{ of actual antenna}}{U \text{ of isotropic antenna}} = \frac{U}{\frac{P_{\text{rad}}}{4\pi}} = \frac{4\pi U}{P_{\text{rad}}}$$

When direction isn't specified, it means that direction of max radiation is implied

$$D_{\text{max}} = \frac{U_{\text{max}}}{U_0} = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}}$$

→ Relation b/w solid angle (Ω) & plane angle (θ, ϕ)

$$r^2 d\Omega = da = r^2 \sin\phi d\theta d\phi$$

$$d\Omega = \sin\phi d\theta d\phi$$

→ $D = \frac{3}{2} \sin^2\theta$ for Hertzian dipole antenna

Maximum directivity of a Hertzian dipole, $D_{\text{max}} = \frac{3}{2} @ \pm \frac{\pi}{2}$

$$D_{\text{max}} (\text{dB}) = 1.76 \text{ dB}$$

→ $D = 1.642 \cdot \left(\frac{\cos(\frac{\pi}{2} \cos\theta)}{\sin\theta} \right)^2$ for $\frac{\lambda}{2}$ dipole antenna

Maximum directivity of a $\frac{\lambda}{2}$ dipole, $D_{\text{max}} = 1.642 @ \frac{\pi}{2}$

$$D_{\text{max}} (\text{dB}) = 2.153 \text{ dB}$$

vii) **Antenna Efficiency**

$$\rightarrow P_{\text{in}} = P_{\text{rad}} + P_{\text{loss}}$$

$$\text{efficiency, } e = \frac{P_{\text{rad}}}{P_{\text{in}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$$

→ Gain of an antenna

$$G = e D$$

$$= \frac{P_{\text{rad}}}{P_{\text{in}}} \cdot \frac{4\pi U}{P_{\text{rad}}} = \frac{4\pi U}{P_{\text{in}}}$$

For zero loss, $e = 1 \Rightarrow G = D \Rightarrow P_{\text{rad}} = P_{\text{in}}$

viii) Field Regions

a) Bandwidth of an antenna

$$BW = \frac{f_u - f_l}{f_0}$$

f_u to $f_l \Rightarrow$ radiation characteristics freq. range

f_0 : centre freq / freq at which radiation characteristics are max

b) Field Regions - Based on distance :

- Reactive Near field region $r < 0.62 \sqrt{\frac{D^3}{\lambda}}$
- Intermediate field region (Fresnel) $0.62 \sqrt{\frac{D^3}{\lambda}} < r < \frac{2D^2}{\lambda}$
- Far field region (Fraunhofer) $r > \frac{2D^2}{\lambda}$

- For wire antennas, D is the largest dimension of antenna

c) Effective aperture of an antenna - Area over which antenna collects electromagnetic energy from incident wave & deliver to load

$$A_e = \frac{P_r}{W_{av}} \Rightarrow P_r = A_e \cdot W_{av} \text{ Watts}$$

$$\left(A_e = \frac{G \lambda^2}{4\pi} \text{ m}^2 \right)$$

For hertzian dipole $\Rightarrow G = D$

$$D = \frac{4\pi U}{P_{rad}} = \frac{3}{2} \sin^2 \theta$$

$$\text{if } \theta = \frac{\pi}{2} \Rightarrow D = \frac{3}{2}$$

$$\text{Radiation resistance, } R_{rad} = 80\pi^2 \times \left(\frac{l}{\lambda}\right)^2$$

$$R_{rad} \times D = \frac{3}{2} \times 80\pi^2 \left(\frac{l}{\lambda}\right)^2 \Rightarrow R_{rad} = \frac{120\pi^2}{D} \left(\frac{l}{\lambda}\right)^2$$

$$\text{Avg. Power Density, } W_{av} = \frac{1}{2} |E| |H| = \frac{1}{2} \frac{|E|^2}{\eta}$$

$$\text{Power delivered to load, } P_r = \frac{|V_a|^2}{8 R_{rad}}$$

$$A_e = \frac{P_r}{W_{av}} = \frac{|V_a|^2}{8 R_{rad}} \times \frac{2\eta}{|E|^2} = \frac{|E|^2 \cdot l^2 \times 2\eta}{8 R_{rad} |E|^2} = \frac{\eta l^2}{4 R_{rad}} = \frac{\eta l^2}{4 \times \left(\frac{120\pi^2}{D}\right) \left(\frac{l}{\lambda}\right)^2} = \frac{\lambda^2 D}{4\pi} \quad \text{For lossless case, } G=D \Rightarrow A_e = \frac{\lambda^2}{4\pi} \cdot G$$

d) Radiation Resistance of an antenna

$$\rightarrow P_{rad} = R_{rad} \times I_{eff}^2$$

$\downarrow$ $\downarrow$ $\downarrow$
 Power Radiated Resistance RMS of current

$$i(t) = \text{Re} [I e^{j\omega t}] = I \cos(\omega t)$$

$$I_{eff} = \frac{I}{\sqrt{2}} \Rightarrow I = \sqrt{2} I_{eff}$$

Any length (other than $\lambda/2$) dipole antenna

$\rightarrow$ Current distribution eqⁿ

$$I(z') = I_0 \sin \left[k \left(\frac{\lambda}{2} - z' \right) \right], \quad 0 \text{ to } \frac{\lambda}{2}$$

$$\left\{ I(z') \right\}_{-\frac{\lambda}{2} \text{ to } 0} = \text{mirror image of } \left\{ I(z') \right\}_{0 \text{ to } \frac{\lambda}{2}}$$

So,

$$A_2 = \frac{\mu}{4\pi} \int_{-\frac{\lambda}{2}}^{\frac{\lambda}{2}} I(z') dz' \frac{e^{-jkR}}{R}$$

$$= \frac{\mu I_0 e^{-jkR}}{2\pi k R} \left[\frac{\cos\left(\frac{k\lambda}{2} \cos\theta\right) - \cos\left(\frac{k\lambda}{2}\right)}{\sin^2\theta} \right]$$

$\rightarrow$ Fields due to dipole of any length

$$H_{\phi_{rad}} = j \frac{I_0 e^{-jkR}}{2\pi R} \left[\frac{\cos\left(\frac{k\lambda}{2} \cos\theta\right) - \cos\left(\frac{k\lambda}{2}\right)}{\sin\theta} \right]$$

$$E_{\theta_{rad}} = \eta H_{\phi_{rad}} = j \eta \frac{I_0 e^{-jkR}}{2\pi R} \left[\frac{\cos\left(\frac{k\lambda}{2} \cos\theta\right) - \cos\left(\frac{k\lambda}{2}\right)}{\sin\theta} \right]$$

$\rightarrow$ Radiated Power Density

$$W_{av} = \frac{1}{2} |E_{\theta_{rad}}| |H_{\phi_{rad}}| = \frac{\eta}{2} |H_{\phi_{rad}}|^2 \text{ W/m}^2$$

$\rightarrow$ Total radiated power

$$P_{rad} = \oint W_{av} \cdot da$$

$$= \oint W_{av} \cdot r^2 \sin\theta d\theta d\phi = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} W_{av} \cdot r^2 \sin\theta d\theta d\phi$$

$$= \frac{\eta I_0^2}{4\pi} \int_0^{\pi} \frac{1}{\sin\theta} \left[\cos\left(\frac{k\lambda}{2} \cos\theta\right) - \cos\left(\frac{k\lambda}{2}\right) \right]^2 d\theta$$

U4 Friis Transmission Formula & Antenna Arrays

Friis Transmission Equation

→ The fundamental eqⁿ which helps determine ratio of received power to transmitted power

D_e : Directivity of T_x antenna at R_x antenna terminals

U_e : Radiation intensity due to T_x antenna @ receiver antenna

$$D_e = \frac{U_e}{U_0} = \frac{U_e}{P_t / 4\pi} = \frac{4\pi U_e}{P_t}$$

For lossless T_x antenna, $D_e = G_t$

$$G_t = \frac{4\pi U_e}{P_t}$$

If S_e is power density due to T_x antenna @ R_x antenna terminals

$$U_e = S_e \cdot r^2$$

$$G_t = \frac{4\pi \cdot S_e r^2}{P_t} \Rightarrow S_e = \frac{G_t P_t}{4\pi r^2}$$

Affective Aperture of R_x antenna, $A_{e_r} = \frac{P_r}{W_{av_e}} \cdot \frac{P_t}{S_e} \Rightarrow S_e = \frac{P_r}{A_{e_r}}$

$$G_t = \frac{4\pi r^2 P_r}{P_t \cdot A_{e_r}}$$

$$P_r = \frac{P_t G_t A_{e_r}}{4\pi r^2}$$

$$A_{e_r} = \left(\frac{\lambda^2}{4\pi} \right) G_r$$

$$\text{So, } P_r = \frac{P_t G_t}{4\pi r^2} \cdot \frac{\lambda^2}{4\pi} \cdot G_r = P_t G_t G_r \cdot \left(\frac{\lambda}{4\pi r} \right)^2$$

$$\text{In dB, } 10 \log_{10} P_r = 10 \log_{10} P_t + 10 \log_{10} G_t + 10 \log_{10} G_r + 20 \log_{10} \frac{\lambda}{4\pi r}$$

$$P_r(\text{dB}) = P_t(\text{dB}) + G_t(\text{dB}) + G_r(\text{dB}) + 20 \log_{10} \frac{\lambda}{4\pi r}$$

Expression for Electric field due to an array of Hertzian dipole placed along z-axis

→ Distance electric field due to single z-directed Hertzian dipole at origin

$$E_{\theta_{\text{rad}}} = \eta H_{\phi_{\text{rad}}} = \frac{j\eta K I l \sin\theta}{4\pi r} e^{-jk r}$$

→ If placed at z'_1 from origin

$$E_{\theta_{\text{rad}_1}} = \frac{j\eta K I l \sin\theta}{4\pi r_1} e^{-jk r_1}$$

for far field region, $r_1 \approx r - z'_1 \cos\theta$

$$E_{\theta_{\text{rad}_1}} = \frac{j\eta K I l \sin\theta}{4\pi r} e^{-jk(r - z'_1 \cos\theta)}$$

$$E_{\theta_{\text{rad}_2}} = \frac{j\eta K I_2 l \sin\theta}{4\pi r} e^{-jk(r - z'_2 \cos\theta)}$$

$$E_{\theta_{\text{rad}}} = E_{\theta_{\text{rad}_1}} + E_{\theta_{\text{rad}_2}} + \dots + E_{\theta_{\text{rad}_N}} = \frac{j\eta K l \sin\theta}{4\pi r} e^{-jk r} \left[I_1 e^{jk z'_1 \cos\theta} + I_2 e^{jk z'_2 \cos\theta} + \dots + I_N e^{jk z'_N \cos\theta} \right]$$

Array Factor

Radiation pattern: Normalized magnitude of Electric Field

Array pattern: Radiation pattern of an array

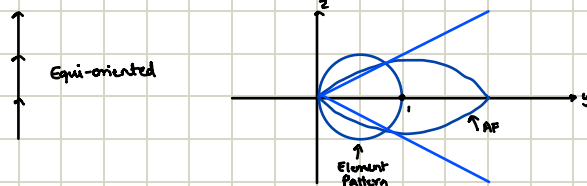
$$\begin{aligned} &= |E_{\theta_{\text{rad}}}|_{\text{nor}} \\ &= |\sin\theta| \left| \sum_{n=1}^N I_n e^{jk z'_n \cos\theta} \right|_{\text{nor}} \\ &= \text{Element pattern} \times \text{AF} \end{aligned}$$

Pattern Multiplication Theorem

→ Radiation pattern of an array of equi-oriented identical antenna elements is product of element pattern & Array factor

$$AF = \sum_{n=1}^N I_n e^{jkz_n' \cos \theta}$$

Simplest array $N=2$



2-Element Array

$$AF = \sum_{n=1}^2 I_n e^{jkz_n' \cos \theta} = I_1 e^{-jk\frac{d}{2} \cos \theta} + I_2 e^{jk\frac{d}{2} \cos \theta} = 2I_0 \cos\left(\frac{\alpha}{2} + \frac{kd}{2} \cos \theta\right)$$

→ There exists phase difference α

i) $\alpha = 0 \Rightarrow$ Phase diff. b/w currents is zero

$$AF = 2I_0 \cos\left(\frac{kd}{2} \cos \theta\right)$$

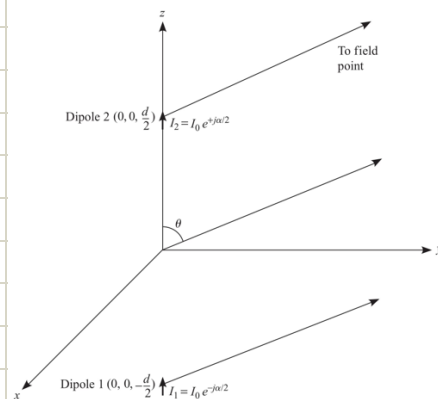
Maxima of 2-element array: $|AF|$ has maxima when $\frac{kd}{2} \cos \theta_m = \pm m\pi$

$$\cos \theta_m = \frac{\pm 2m\pi}{kd} = \pm \frac{2m\pi}{\frac{2\pi d}{\lambda}} = \pm \frac{m\lambda}{d}$$

$$|\cos \theta_m| \leq 1 \Rightarrow \frac{m\lambda}{d} \leq 1$$

$$m \leq \frac{d}{\lambda} \Rightarrow \text{Maxima}$$

P	0	1	2
d	$0 \leq d < \lambda$	$\lambda \leq d < 2\lambda$	$2\lambda \leq d < 3\lambda$
m	0	0 1	0 1 2
$\cos \theta_m$	0	0 $\pm \lambda/d$	0 $\pm \lambda/d$ $\pm 2\lambda/d$
θ_m	$\pi/2$	$\pi/2$ $\cos^{-1}(\pm \lambda/d)$	$\pi/2$ $\cos^{-1}(\pm \lambda/d)$ $\cos^{-1}(\pm 2\lambda/d)$



Increase in element spacing leads to narrowing of main beam & appearance of more lobes

No. of maxima in AF depends on spacing b/w elements

If spacing is less than wavelength, only 1 maxima in AF

AF has maximum along $\theta = 90^\circ$ (i.e.) **broadside** $\Rightarrow$ This array is called **Broadside array**

Nulls of 2-element array ($\alpha = 0$) (Broadside Array)

$$\rightarrow AF = 0 = 2I_0 \cos\left(\frac{kd}{2} \cos \theta\right) \Rightarrow \frac{kd}{2} \cos \theta = \pm \frac{(2n-1)\pi}{2} \Rightarrow \frac{d}{\lambda} \cos \theta_n = \pm \frac{(2n-1)}{2} \Rightarrow \theta_n = \cos^{-1}\left[\pm \frac{(2n-1)}{2} \cdot \frac{\lambda}{d}\right]$$

$$\text{For result to be on RHS, } \frac{2n-1}{2} \cdot \frac{\lambda}{d} \leq 1 \Rightarrow n \leq \frac{d}{\lambda} + \frac{1}{2}$$

2-Element array with non-zero phase shift

$$\rightarrow \alpha \neq 0 \Rightarrow AF = 2I_0 \cos\left(\frac{\alpha}{2} + \frac{kd}{2} \cos \theta\right)$$

$$\text{Let } \alpha = kd, AF = 2I_0 \cos\left(\frac{kd}{2} (1 + \cos \theta)\right)$$

a) Maxima when $\alpha = kd$ for end fire array

$$\frac{kd}{2} (1 + \cos \theta) = \pm m\pi$$

$$\frac{2\pi}{\lambda} \times \frac{d}{2} (1 + \cos \theta) = \pm m\pi \Rightarrow 1 + \cos \theta_m = \pm \frac{m\lambda}{d} \Rightarrow \cos \theta_m = \pm \frac{m\lambda}{d} - 1$$

$$\text{If } m=0, \cos \theta_m = 0 - 1 = -1 \Rightarrow \theta_m = \pi \Rightarrow \text{maxima}$$

$\hookrightarrow$ Hence, **End-fire Array**

b) Nulls of End fire Array

$$n \leq \frac{2d}{\lambda} + \frac{1}{2}$$

$$\cos \theta_n = \left(\frac{2n-1}{2}\right) \frac{\lambda}{d} - 1 \quad (\text{Similar steps as broadside})$$

$$\begin{aligned} \cos \theta_m &= \frac{m\lambda}{d} - 1 \Rightarrow \text{valid when } 0 \leq \frac{m\lambda}{d} \leq 2 \quad \checkmark \\ \cos \theta_m &= -\frac{m\lambda}{d} - 1 \Rightarrow \text{valid when } 0 \geq \frac{m\lambda}{d} \geq -2 \\ &\text{But } \frac{m\lambda}{d} \text{ always +ve, so } \times \end{aligned}$$

Uniform Array of N-elements

$$\rightarrow AF = \sum_{n=1}^N I_n e^{jkz'_n \cos \theta}$$

$$AF = I_0 + I_0 e^{j\kappa} e^{j\kappa d \cos \theta} + I_0 e^{j2\kappa} e^{j\kappa (2d) \cos \theta} + \dots$$

$$= I_0 [1 + e^{j\kappa(1 + d \cos \theta)} + e^{j\kappa(2 + 2d \cos \theta)} + \dots]$$

$$\text{Let } \alpha + \kappa d \cos \theta = \psi$$

$$AF = I_0 [1 + e^{j\psi} + e^{j2\psi} + \dots + e^{j(N-1)\psi}]$$

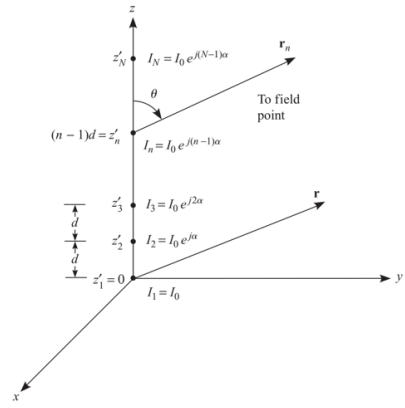
$$AF \cdot e^{j\psi} = I_0 [e^{j\psi} + e^{j2\psi} + \dots + e^{jN\psi}]$$

$$AF(e^{j\psi} - 1) = I_0 [e^{jN\psi} - 1]$$

$$AF = I_0 \cdot \frac{e^{jN\psi} - 1}{e^{j\psi} - 1}$$

$$AF = I_0 e^{j(N-1)\psi/2} \cdot \frac{\sin \frac{N\psi}{2}}{\sin \frac{\psi}{2}}$$

Element	Current	Distance
1	$I_1 = I_0$	$z'_1 = 0$
2	$I_2 = I_0 e^{j\kappa}$	$z'_2 = d$
3	$I_3 = I_0 e^{j2\kappa}$	$z'_3 = 2d$
$\vdots$	$\vdots$	$\vdots$
N	$I_N = I_0 e^{jN\kappa}$	$z'_N = (N-1)d$



Magnitude of AF of N-elements

$$\rightarrow |AF|_n = \frac{|AF|}{|AF|_{\max}} = \frac{I_0 \sin \left(\frac{N\psi}{2} \right)}{\sin \left(\frac{\psi}{2} \right)} = \frac{\sin \left(\frac{N\psi}{2} \right)}{N \sin \left(\frac{\psi}{2} \right)}$$

Maxima of |AF|_n of N-elements Uniform array

$\rightarrow$ To find maxima, $\frac{\partial}{\partial \psi} (|AF|_n) = 0$ & find ψ

$$\frac{\partial}{\partial \psi} (|AF|_n) = \frac{\partial}{\partial \psi} \left(\frac{\sin \left(\frac{N\psi}{2} \right)}{N \sin \left(\frac{\psi}{2} \right)} \right) = \frac{1}{N} \cdot \left(\frac{\sin \frac{\psi}{2} \cdot \frac{N}{2} \cos \frac{N\psi}{2} - \sin \frac{N\psi}{2} \cdot \frac{1}{2} \cos \frac{\psi}{2}}{\sin^2 \frac{\psi}{2}} \right) = 0 \Rightarrow N \sin \left(\frac{\psi}{2} \right) \cos \left(\frac{N\psi}{2} \right) = \sin \left(\frac{N\psi}{2} \right) \cos \left(\frac{\psi}{2} \right) \Rightarrow N \tan \frac{\psi}{2} = \tan \left(\frac{N\psi}{2} \right)$$

$$\text{Then, } \frac{\psi}{2} = \pm m\pi \Rightarrow \psi = \pm 2m\pi$$

$\rightarrow$ AF has periodic maxima at intervals of 2π . Principal maxima at $m=0$ called **Main lobe**

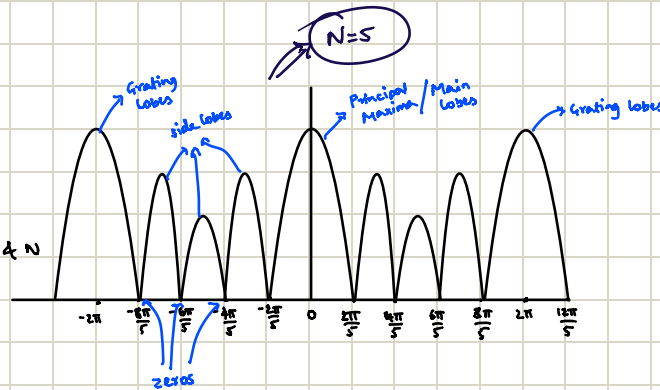
All other lobes containing maxima are called **Grating lobes**

Nulls of N-element Uniform Array

$$\rightarrow |AF|_n = \frac{\sin \left(\frac{N\psi}{2} \right)}{N \sin \left(\frac{\psi}{2} \right)}$$

$$\text{Nulls occur at } \frac{N\psi}{2} = \pm p\pi \Rightarrow \psi_p = \pm \frac{2p\pi}{N}, \quad p = 1, 2, \dots, N-1$$

at $p=0$ & N , denominator will be zero, so no nulls at $p=0$ & N



$\rightarrow$ **Main lobe** - Largest & Strongest radiation beam

- Points in desired direction of transmission

$\rightarrow$ **Side lobe** - Smaller lobes on either sides of main lobe

- Represent undesired radiation in other directions

- Usually lower in magnitude than main lobe

$\rightarrow$ **Grating lobe** - Large unwanted lobes (sometimes equal to main lobe)

- Occur when element spacing $d > \lambda/2$

- Can radiate power in completely wrong directions

Array Factor of an array with non-uniform excitation

$$\begin{aligned} \rightarrow AF &= I_0 + I_1 e^{j\psi} + I_2 e^{j2\psi} + \dots + I_{N-1} e^{j(N-1)\psi} \\ &= I_0 \left[1 + \frac{I_1}{I_0} z + \frac{I_2}{I_0} z^2 + \dots + \frac{I_{N-1}}{I_0} z^{N-1} \right] \quad (e^{j\psi} = z) \\ &= I_0 [1 + a_1 z + a_2 z^2 + \dots + a_{N-1} z^{N-1}] \end{aligned}$$

Binomial array of N elements

$$AF = I_0 [1 + z]^{N-1}$$

$$AF_3 = I_0 [1 + z]^2 = I_0 [1 + 2z + z^2]$$

$$AF_4 = I_0 [1 + z]^3 = I_0 [1 + 3z + 3z^2 + z^3]$$

$$AF_5 = I_0 [1 + z]^4 = I_0 [1 + 4z + 6z^2 + 4z^3 + z^4]$$

Pascal's Triangle:

1					
2	1	1			
3		1	2	1	
4			1	3	3
5				1	4

Nulls of binomial array of N elements

$$\rightarrow |AF|_n = (1 + z)^{N-1} = 0$$

$$\psi_n = \pm (2n-1)\pi$$

$$\text{But } \psi_n = Kd \cos \theta_n + \alpha$$

$$\& \alpha = 0$$

$$\text{So, } \pm (2n-1)\pi = Kd \cos \theta_n$$

$$\cos \theta_n = \frac{\pm (2n-1)\pi}{\frac{2\pi}{\lambda} \cdot d} = \frac{\pm (2n-1)(\lambda/2)}{d}, \quad n=1, 2, \dots$$

We want on RHS, so only +

$$\cos \theta_n = \frac{(2n-1)}{d} \cdot \frac{\lambda}{2}$$

Visible Region

$$\rightarrow \psi = Kd \cos \theta + \alpha$$

θ varies from 0° through 90° to 180° . ψ takes values $Kd + \alpha$ through α to $-Kd + \alpha$

$$Kd + \alpha \geq \psi \geq -Kd + \alpha \Rightarrow \text{Visible Region}$$

$\rightarrow$ As element spacing increases, no. of lobes in VR increases

Broadside Array

$\rightarrow$ Desirable requirement: Max radiation of an array directed normal to axis of array ($\theta = 90^\circ$)

$\rightarrow$ Design: $\psi = Kd \cos \theta + \alpha = 0$

First max directed towards $\theta = 90^\circ$

$$\psi = Kd \cos \theta + \alpha \Big|_{\theta=90^\circ} = \alpha = 0$$

To have max of the AF of uniform linear array directed to axis of array, all elements should have phase excitation

To ensure no principal max in other directions, separations shouldn't be equal to multiples of wavelength ($d \neq n\lambda$, $n=1, 2, 3, \dots$)

$$\text{If } d = n\lambda, \quad n=1, 2, 3, \dots \& \alpha = 0, \text{ then } \psi = Kd \cos \theta + \alpha \Big|_{\substack{d=n\lambda \\ \alpha=0 \\ n=1, 2, 3, \dots}} = 2\pi n \cos \theta \Big|_{\theta=0^\circ, 180^\circ} = \pm 2\pi n$$

To avoid any grating lobe, largest spacing should be less than one wavelength ($d_{\max} < \lambda$)

Ordinary Endfire Array

$\rightarrow$ Instead of having max radiation broadside to axis of array, we desire to have along axis array (Endfire array) $\theta = 0^\circ, 180^\circ$

$\rightarrow$ To direct first max towards $\theta_0 = 0^\circ$, $\psi = Kd \cos \theta + \alpha \Big|_{\theta=0^\circ} = Kd + \alpha = 0 \Rightarrow \alpha = -Kd$

If max is desired towards $\theta = 180^\circ$, $\psi = Kd \cos \theta + \alpha \Big|_{\theta=180^\circ} = -Kd + \alpha = 0 \Rightarrow \alpha = Kd$

$\rightarrow$ If element separation is $d = \frac{\lambda}{2}$

$\hookrightarrow$ Endfire in both directions ($\theta = 0^\circ \& 180^\circ$) simultaneously

If element spacing is multiple of wavelength ($d = n\lambda$)

$\hookrightarrow$ There exists maxima in broadside directions

$\rightarrow$ To have only 1 max & no grating lobes $\Rightarrow$ max spacing, $d_{\max} < \frac{\lambda}{2}$

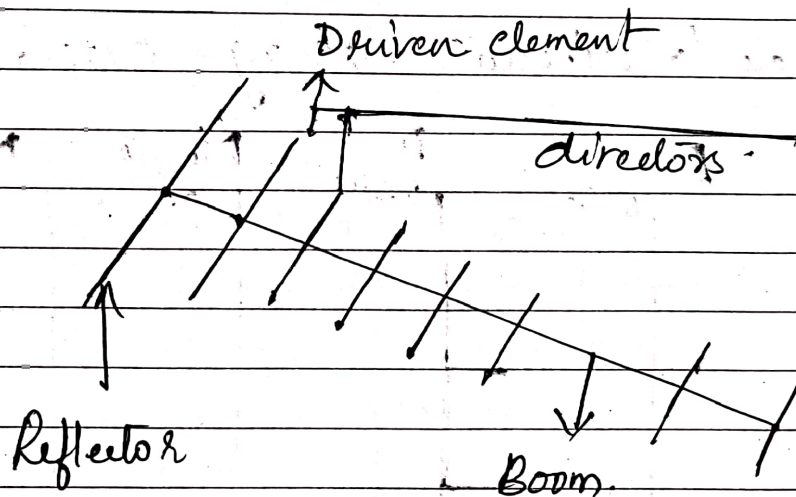
Special Antennas:-

1. Yagi Uda antenna:-

It is a directional antenna invented by Yagi and Uda. It is designed to enhance signal strength in a single direction, making it ideal for long distance communication or areas with weak signals.

Key Components are

- 1) Driven element that produces electromagnetic field. (Primary element)
- 2) Reflector that reflects energy forward to strengthen the signal.
- 3) Directors to focus and direct the signal to minimize interference.



Primary element is driven element responsible for TX and reception.

At the far end of the antenna is the reflector a longer element that reflects energy towards the antenna's radiation pattern.

Directors:- In front of the driven element, there are multiple strategically placed directors with specific lengths and spacing.

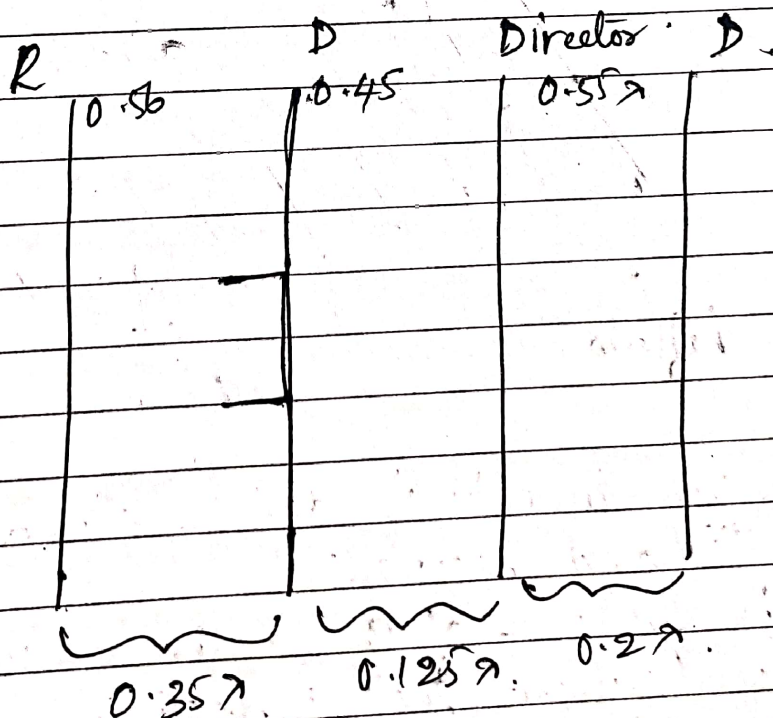
These directors assist in directing the antenna's radiation pattern for optimal performance.

Driven element	—	0.458λ to 0.5λ
Reflector	—	0.55λ to 0.58λ
Directors	—	1 0.45λ
		2 0.40λ
		3 0.35λ

Spacing between directors — 0.2λ

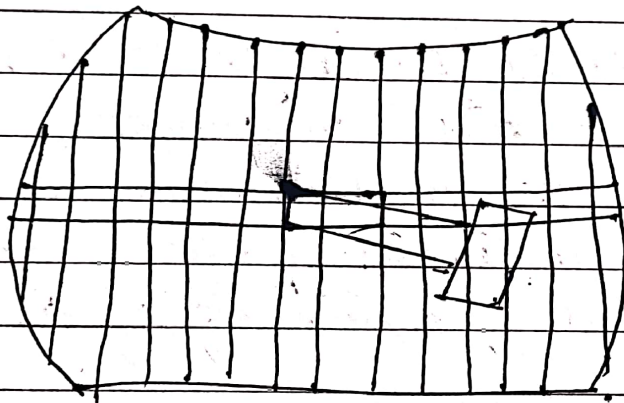
Reflector to dipole spacing — 0.35λ

Dipole to director spacing — 0.125λ



Applications:-

1. Yagi Uda antenna improves TV signal reception in areas with weak or distant broadcasting stations.
2. Amateur radio operators help Yagi antennas for efficient point to point comm with their directional capabilities.
3. Yagi antennas are applied in point to point wireless links, extending the range and reliability of comm. links.



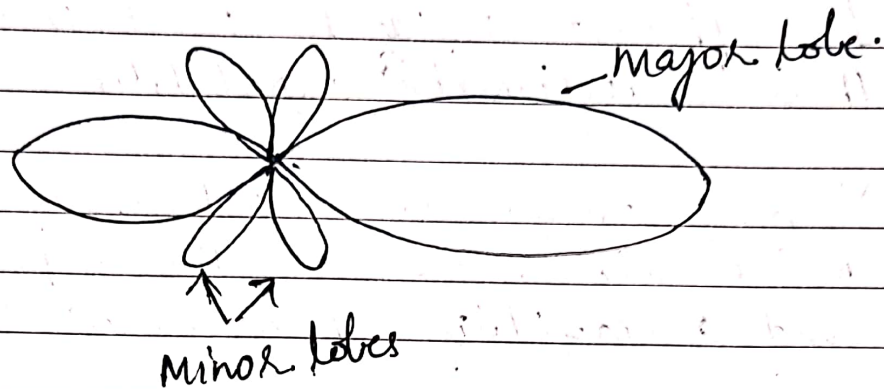
Reflector — Parabolic Reflector is a curved surface that collects and focuses incoming em waves onto the feed antenna.

Yagi Uda — It is placed at the focal point of the parabolic reflector that receives the focused em waves.

Signal amplification:- Yagi antenna amplifies the received signal and then TXed to RX or TXes.

Radiation Pattern of Yagi-Uda.

The lobe corresponds to the direction in which the antenna is precisely aimed making it highly effective in that direction.



Can customise the length and spacing of its reflector and other components.
Can reduce the radiation in minor lobes, or amplify antennas forward gain.
No universal solution for all scenarios.

Advantages

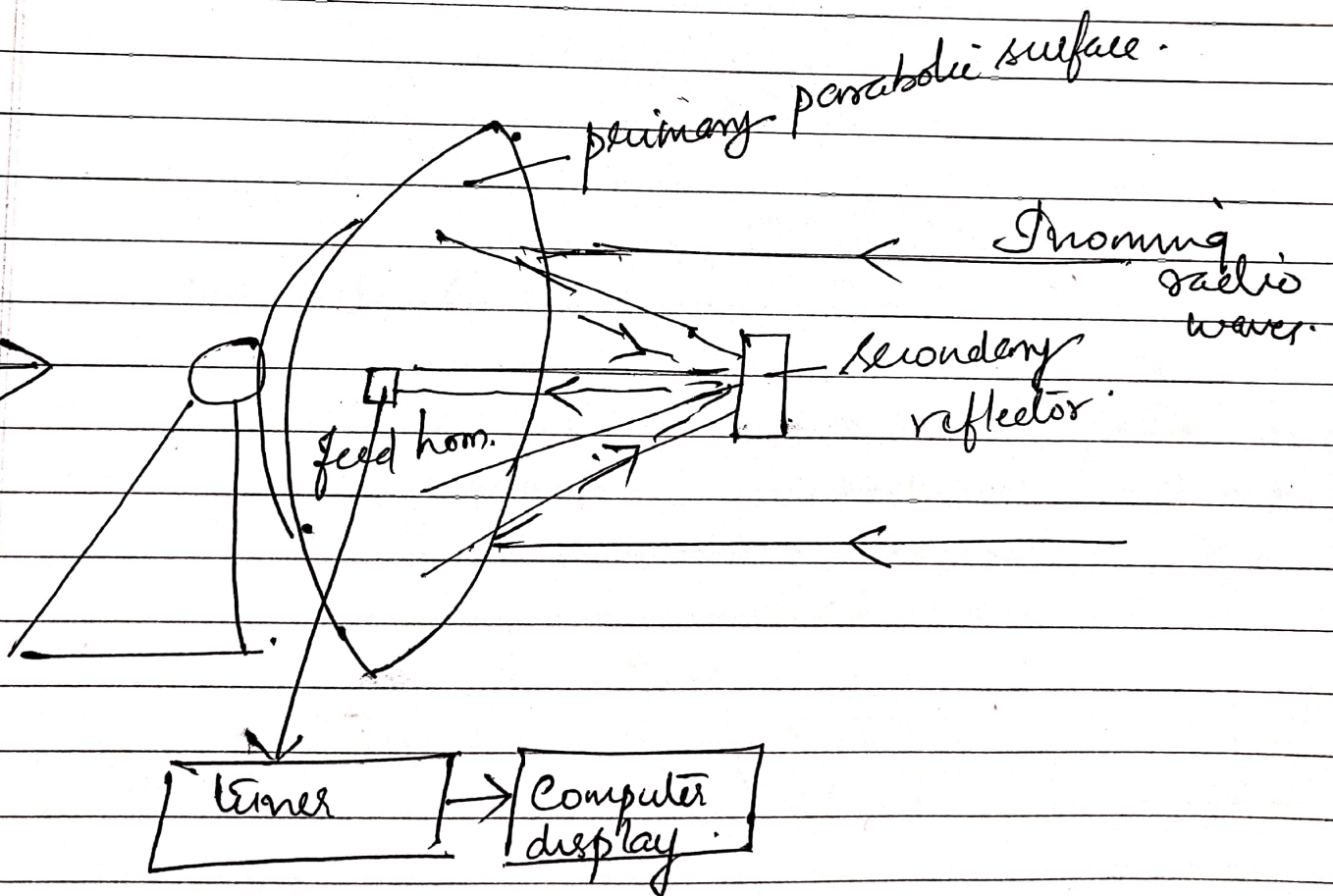
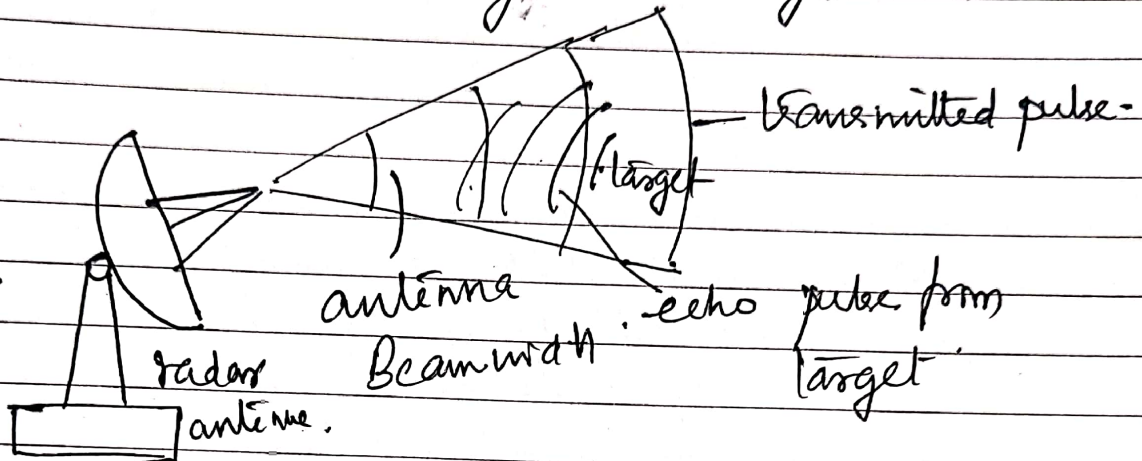
Improved signal strength - point signals in one direction reducing interference.
Better reception.
Make signals stronger

Disadvantages :-

Limited freq range :- Work best within a small range of frequency.
Size constraint :- Long Yagi antennas can be too long when space is tight.

In radar installations, Yagi Uda antennas are chosen for their directional properties, that helps in precise target detection and tracking.

Radio telescopes use Yagi antennas to collect and analyse cosmic radio waves from specific celestial objects or regions.



Yagi Uda antennas are suitable for satellite dish installations, aiding in signal reception from orbiting satellites.

Employed in scenarios where multiple locations need to communicate with a central point such as remote sensing or surveillance systems.

Helical antenna.

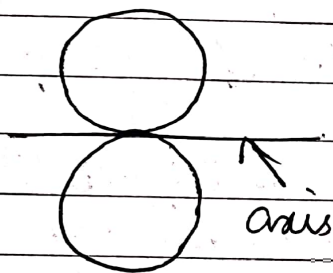
It is a type of radio antenna that consists of a helix (a spiral shaped conductor) that is used to transmit and receive ~~the~~ em waves. D, S, N determine fr, gain, BW.

These antennas operate in two primary modes

normal (broadside mode)
axial (end fire).

(pitch and diameter)

Normal mode have dimensions small compared to the wavelength and axial mode have dimensions similar to the wavelength.



Normal $C < \lambda$

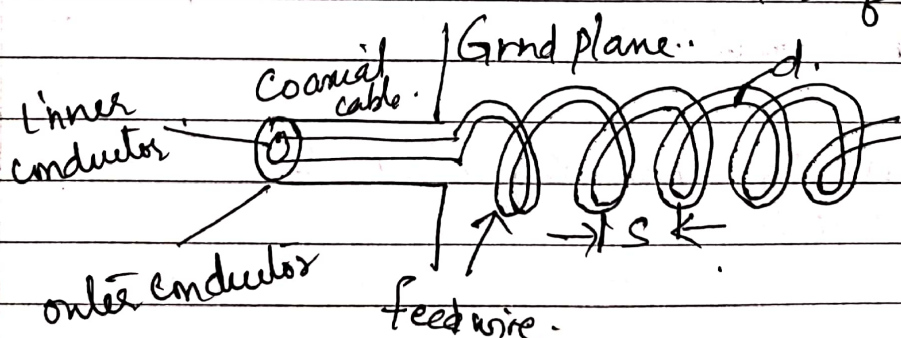


Axial mode

$C = \lambda$

operation is similar to dipole or monopole.

radiation pattern is directional, with max rad intensity along the axis of helix.



Characteristics:-

Circular polarization:- It produces circularly polarized waves, which can reduce signal loss due to polarization mismatch.

End fire radiation:- It radiates energy in the direction of the helix axis.

High gain for long distance comm.

Applications

Used in satellite communication for TXing and receiving signals due to its ability to provide circular polarization which is beneficial for satellite comm.

Circular polarization mitigates the effect of polarization rotation that can occur when signals travel thro the ionosphere.

Wide Bandwidth:- Helical antenna offer a wider bandwidth, i.e. they can transmit and receive a broad range of frequencies which is needed for satellite comm.

Can manage misalignment b/w antenna and RX device.

Satellite tracking :-

Space Probe Comm :- TX or Reception of data between a space probe and earth or other spacecraft.

Space Telemetry

This comm is mainly for Command & Control, sending scientific data, images etc, navigation tracking to find trajectory etc.

Space Probe is an unmanned spacecraft designed to study and space, planets, moons, asteroids, comets and other celestial bodies.

It has various equipments or instruments and sensors to collect data, conduct exp and gather info about target environment.

DSN - Deep space Net - Net of antennas around the world that communicate with space probes.

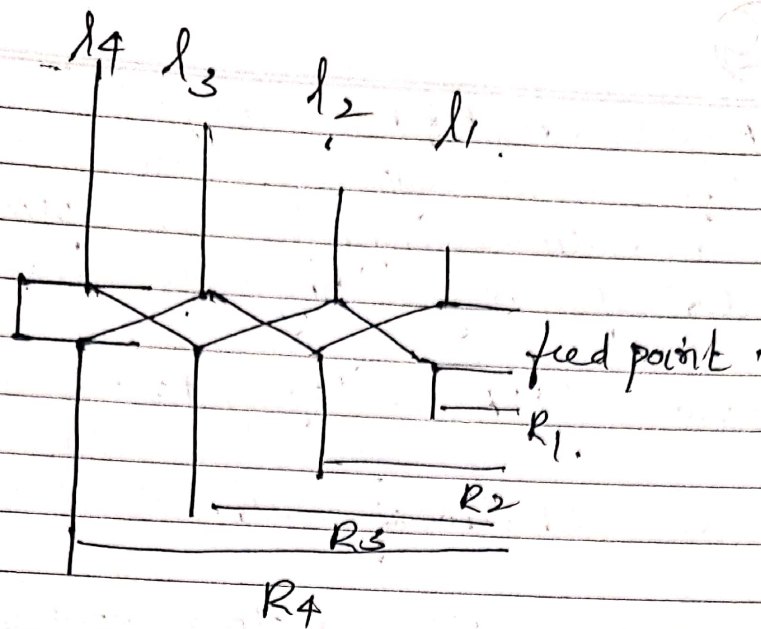
Weather and Data Relay Satellite :-

TX & RX data from weather satellites.

Relay of data b/w satellites and ground stations.

Space Telemetry :-

Ability to receive VHF signals



The dipole lengths and separations are related

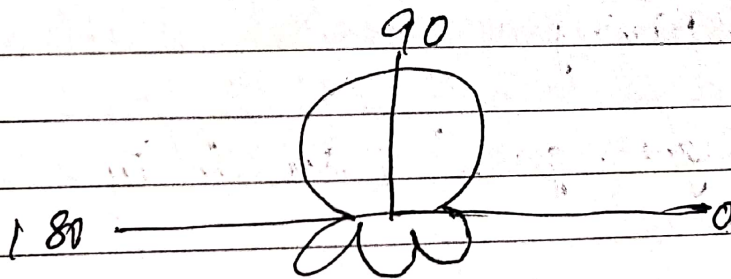
$$\frac{R_1}{R_2} = \frac{R_2}{R_3} = \frac{R_3}{R_4} = \frac{l_1}{l_2} = \frac{l_2}{l_3} = \frac{l_3}{l_4} = T < 1$$

R - distance btw feed & dipole

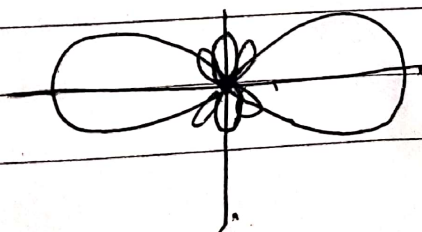
l - length of dipole

The radiation pattern may be unidirectional or bidirectional.

For unidirectional LP antenna



Bidirectional LP antenna, max radiation is in Broadside



* LPDA is a backfire array.

Backfire is a type of antenna array that radiates energy in the opposite direction of arrays physical orientation.

Compare LPDA & Yagi Uda.

* Both are directional antennas.

* Both designed to radiate energy in specific direction.

* Both antennas use multiple elements to achieve directivity and gain.

Diff

operates over

LPDA ~~freq~~ wide freq range.

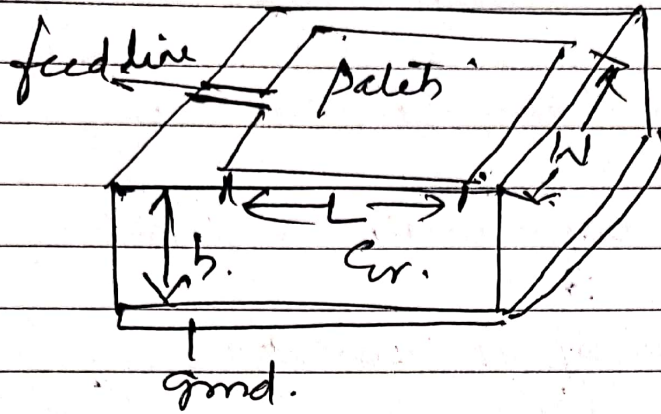
Yagi designed for specific range.

LPDA has log periodic structure with elements of varying lengths.

Yagi Uda antennas have fixed element length and spacing.

Yagi has higher gain & directivity.

Microstrip Patch antenna.



The patch aids in signal shaping and T^x ion whereas ground plane is responsible for increasing the effectiveness of entire antenna by providing a stable reference point.

3 types

Microstrip patch

" slot

Printed dipole antenna